# Solution Requirements

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| **Date** | 1 July 2026 |
| **Team ID** | |
| **Project Name** | **Human Development Index (HDI) Predictor – A Comprehensive Measure of Well-Being** |
| **Maximum Marks** | 4 Marks |

**Define Functional and Non-Functional Requirements:**

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category | Requirement Description | Priority (High/ Medium/Low) |
|---|---|---|---|
| 1 | Authentication | *The system allows authorized users (Researchers/Policymakers) to access the HDI prediction application. User login can be added in future versions if required.* | Medium |
| 2 | Authorization levels | All authorized users have permission to enter development indicators and view HDI prediction results. | Medium |
| 3 | External interfaces | The system reads the **HDI.csv** dataset and loads the trained **Random Forest model (hdi_model.pkl)** | High |

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| | | *and label_encoder.pkl for prediction.* | |
| **4** | Transactions processing | The application accepts Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita, validates the values, processes them through the trained model, and returns the predicted HDI category instantly. | High |
| **5** | Reporting | *The system displays the predicted HDI category (Very High, High, Medium, or Low) through the Streamlit interface.* | High |
| **6** | Business rules, etc. | All input values must be valid numeric values within acceptable ranges before prediction. Prediction is performed only after successful validation using the trained Random Forest model. | High |
| **7** | Compliance to laws or regulations | The system is intended for educational and research purposes. No personal or sensitive user information is collected or stored, supporting basic data privacy principles. | Medium |
| **8** | *Other* | The system allows multiple predictions with different country indicator values without restarting the application. | Medium |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|---|---|---|---|
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|---|---|---|---|
| **1** | Performance & Speed | *The application should generate HDI predictions quickly after user input.* | Prediction displayed in **less than 2 seconds** |
| **2** | Scalability | The system should support additional countries, larger datasets, and future model improvements. | Supports future dataset expansion without major code changes |
| **3** | Security & Data Privacy | No user personal information is stored. Input validation prevents invalid data from affecting predictions. | Safe handling of user input with no persistent storage |
| **4** | Reliability & Availabilit y | *The application should consistently provide predictions whenever the trained model is available.* | **99% availability** during normal operation |
| **5** | Usability & Accessibility | The Streamlit interface should be simple, user-friendly, and easy to understand for researchers and policymakers. | Users can complete prediction in **less than one minute** |
| **6** | *Other* | The application should be easy to maintain, update, and deploy on different operating systems supporting Python. | Compatible with **Windows, macOS, and Linux** using Python 3 and Streamlit |